Figure 1: Security in the cloud

in a single apartment building, where every tenant has his/her own set of keys to enter his/her flat. No one can step into someone else's flat, unless the door is kept open or the tenant forgets to lock it. This is how security in shared infrastructure works. By managing proper authentication, one can reduce the chance of a security breach considerably. But customer awareness is necessary in this respect.

- **Dedicated infrastructure**: This is when customers are allotted dedicated machines and not shared resources. Here you have the entire apartment building to yourself, where you have the authority to appoint a guard to take care of it or do it on your own. Since you have rented the entire apartment building, you need to pay a high cost. The cloud providers only interfere if you ask them to do so. You can provide any level of security you wish or can afford to.

From the hardware point of view, you are guaranteed high reliability, high durability, and high availability. You are covered against disaster as the service providers give you the option of storing your data in multiple zones. It is highly unlikely that data stored in different zones and in different data centres will become unavailable at the same time, even in the case of some natural disaster—as it is not going to occur at all the data centres at the same time.

Cloud providers like Amazon provide identity and access management (IAM). This allows organisations to enforce role based controls on what a user can do in an AWS environment (IAM integrates with users' existing Active Directory or other authentication platforms). Breech of this kind of security depends upon how well you are providing access controls based on the roles you have created for the user(s). The cloud providers offer you the ability to analyse the activity on the cloud through detailed logs, using which you can audit any unusual or potentially malicious activity.

Figure 2 gives an example of the cloud security provided by AWS.

Security in a cloud environment is a shared responsibility. It differs based on which service model and deployment model is used. The security issues fall into two categories:

1. Those faced by cloud service providers
2. Those faced by customers

AWS provides five layers of security, and it would be good if these are followed by the rest of the cloud service providers. The five layers are listed below.

Figure 2: AWS cloud security

Figure 3: AWS IAM

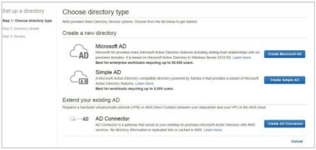

Figure 4: AWS Directory Service

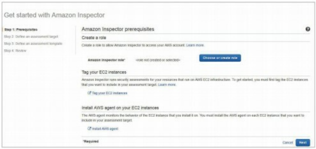

Figure 5: AWS Inspector

Identity and Access Management (IAM): This provides role based authentication.

Directory Service: This provides easy set-up with your existing Microsoft AD, Simple AD and AD connector.

Inspector: Amazon Inspector automatically assesses applications for vulnerabilities or deviations from best practices.

WAF: This is a Web application firewall, which protects